

Structure, Not Magic: An Open Replication and Predictability Ceiling for the Bueno de Mesquita Forecasting Model

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Changes since v5: the reviewer response incorporated. Section 3 reports the full loss-function picture (median AE, hit rates, CRPS, the win split with a sign test, and the mean-AE bootstrap CI) and reframes the result as a bias-for-variance trade rather than a defeat; Section 5 replaces the oracle with a residual probe (cross-validated R^2 spanning zero) and states the benchmark's minimum detectable effect, scoping the ceiling to mean-error losses; the abstract adds the robustness grids and the best-case-configuration point; Section 6 adds an availability-bias account of the legend's persistence; two DEU corrections (capability is exogenous, not a coded column; the 351 issues span all three waves); the OPEC within-question paragraph is cut; a tone pass; new citations (Salganik et al. 2020, Green & Armstrong 2015, Schneider et al. 2010, Tversky & Kahneman 1973); and the provenance footnotes moved to an appendix table.

Abstract

The expected-utility group-decision model of Bueno de Mesquita — deployed by the CIA as Policon and long credited with striking forecasting accuracy — has never had a complete open implementation or a fair public evaluation. We provide both. Our replication, built from the Scholz–Calbert–Smith reconstruction with every interpretive choice logged, reproduces the published reference case. Evaluated on 351 expert-coded EU legislative controversies with sourced actor capabilities and reference points, the model fails a pre-registered accuracy gate against a far simpler benchmark: the capability-and-salience-weighted mean of actor positions. A pre-registered successor search — its data split committed to version control before any fitting — produces two structurally-motivated candidate models; a tournament of four further solution concepts drawn from the field's other traditions extends the comparison to eight in all, counting the weighted mean and the challenge model, and none separates from the mean on held-out data. A flexible learner fitted to the residual the weighted mean leaves behind recovers no signal beyond it, placing the mean at the extractable-signal ceiling for mean-error losses in this domain and input set; and because the 351 issues already combine all three DEU collection waves, the benchmark's minimum detectable effect — about three scale units on a paired mean-absolute-error difference — is a property of the benchmark that more data cannot lower. The robustness is exhaustive on both axes a critic could press: no reading of the ambiguous reconstruction choices, and no configuration of the evaluation, lets the challenge model beat the mean, and the configuration the paper reports is the challenge model's own best case. The one result that does not generalise is reported as such: under a hit-rate loss the challenge dynamics are competitive, a bias-for-variance trade rather than a defeat. We conclude that the lineage's

documented successes, including its models outperforming the very analysts who supplied their inputs, are attributable to structured elicitation rather than to solution dynamics: structure, not magic. The model’s home-turf claim — superiority in coercive settings — remains open, pending an ex-ante coded case library we introduce alongside a live, cryptographically sealed forecast ledger, which has now recorded its first graded cycle. All results regenerate by one command from repository artifacts, given the one third-party dataset the paper documents.

AI use. Both hypothesis generation — two of the eight compared solution concepts were proposed by a large language model — and implementation, the solver, search harness, and scoring code, used AI assistance, in every case under a pre-registered protocol whose train/development/test split and acceptance thresholds were committed to version control before any code or model existed. All data, protocols, execution, and verdicts belong to the deterministic apparatus; the full statement is in the Declarations.

1. Introduction

Few models in political science carry a reputation as outsized, or as strangely unexamined, as the expected-utility group-decision model of Bruce Bueno de Mesquita. Deployed inside the Central Intelligence Agency under the name Policon during the 1980s, it is the subject of a declassified internal evaluation reporting that its forecasts were accurate in roughly ninety percent of applications — and, more strikingly, that the model outperformed the very analysts who supplied its inputs (Feder 1987). Its descendants advised governments and corporations for decades; its inventor was profiled as a “new Nostradamus.” Yet across forty years of influence, no complete public implementation has existed. The model’s full specification was never published. The most serious reconstruction effort — three Australian defence scientists reverse-engineering the equations from scattered fragments — succeeded in reproducing published examples while documenting ambiguities, sensitivity concerns, and convergence problems the source literature never addresses (Scholz, Calbert and Smith 2011). Its commercial successor, Senturion, is proprietary. The result is an epistemic anomaly: one of the most celebrated forecasting instruments in the discipline has never been independently testable. Unfalsifiable famous models are not merely gaps in a literature; they distort it, anchoring expectations about what formal methods can do to claims no one can check.

This paper closes that anomaly, and reports what closing it reveals. We make six contributions. First, we provide an open, deterministic implementation of the model, built strictly from the Scholz reconstruction with every interpretive choice logged, which reproduces the published reference case (9.530). Second, we give the model the fair evaluation it never received: 351 expert-coded legislative controversies from the Decision-Making in the European Union datasets (Thomson et al. 2006), with treaty-regime voting weights supplied as actor capabilities and reference points restored — and the model fails a pre-registered accuracy gate against a radically simpler benchmark, the capability-and-salience-weighted mean of actor positions. Third, we conduct a pre-registered successor search, committing the train/development/test partition to version control before any model fitting, so the repository’s own history certifies that held-out data was never consulted; two structurally motivated candidate models fail on the untouched test split (22.09 vs 21.26, 21.57 vs 21.09), and four further solution concepts drawn from the game-theory literature, scored afterward on the same split, fail alike — no operator we tried beats the weighted mean, and the lone nominal winner’s margin lies inside its own confidence interval once the multiplicity of comparisons is accounted for (Section 4). Fourth, a deliberately flexible cross-validated oracle model scores worse than the weighted mean (-0.84), indicating that the simple benchmark sits at the extractable-signal ceiling for this domain

and input set: the elaborate machinery was not losing to a better model, but to the absence of further signal. Fifth, these results support a reinterpretation of the lineage’s documented successes. If any sane aggregation of structured inputs approaches the ceiling, then Feder’s most puzzling finding — the model beating its own input-providers — requires no special dynamics to explain. The operative technology was never the solution concept; it was the discipline of forcing qualitative expertise into positions, salience, and capability before aggregating. Structure, not magic. Sixth, we introduce the apparatus for the question this paper deliberately leaves open: whether the model’s bargaining dynamics earn their keep in the coercive settings for which they were designed. We contribute a blind dual-entry protocol for building an ex-ante coded case library from the lineage’s own published applications, and a commit-reveal forecast ledger in which live predictions are cryptographically sealed before events resolve (34.576, 41.636, 29.407, 39.443). The coercive-domain verdict is explicitly pending; we report the infrastructure and refuse the temptation.

Two scope guards govern everything that follows, stated here rather than discovered in the limitations section. The predictability-ceiling claim is conditional: it concerns cooperative legislative bargaining as represented in DEU, using the classic input set this lineage has always used — positions, salience, capability, reference points. Richer inputs may hold further signal; coercive domains may reward mechanism. And nothing in this paper evaluates the elicitation step itself, which our reinterpretation identifies as the lineage’s true engine; we take the expert-coded inputs as given. Every number cited in this paper regenerates by one command from repository artifacts, given the one third-party dataset this paper documents, with per-claim provenance; the repository, not the prose, is the source of truth.

The remainder proceeds as follows. Section 2 presents the replication. Section 3 the fair-fight evaluation. Section 4 the successor search. Section 5 the oracle and the ceiling. Section 6 the reinterpretation. Section 7 the case-library protocol. Section 8 the commit-reveal ledger. Section 9 limitations. Section 10 concludes.

2. An open replication of the group-decision model

Our implementation takes as its source not the primary literature — which never contained a complete specification — but the Scholz–Calbert–Smith reconstruction, the only published attempt to assemble the model’s equations in one place. We implement its pipeline in full: effective actor weights from capability and salience; pairwise positional contests and the weighted-median forecast they induce; expected utilities of challenge for every ordered dyad; risk propensities derived from each actor’s security and applied as utility exponents; the classification of dyads into relational octants; the exchange of proposals and consequent position updates; and iteration to convergence. The implementation is deterministic by construction — every stochastic path takes an explicit seed, and identical inputs produce byte-identical outputs — so that the artifact itself, rather than any prose description, serves as the model’s operational specification.

Reconstruction, however, is not transcription. The source literature contains genuine ambiguities, and our first methodological commitment is that every interpretive choice is logged in a public decisions file rather than silently embedded in code. Four examples convey the character of the problem. The security term that drives risk propensity is printed with contradictory superscripts in different parts of the source material; we adopt the reading in which an actor’s security depends on the expected utilities *adversaries* hold for challenging it, on the theoretical ground that risk should derive from exposure. The sign condition defining the compromise octant contradicts its own

accompanying figure; we follow the figure, since compromise is definitionally the both-positive quadrant. The boundaries separating several octants exist only as an unannotated diagram; we implement the natural reading — sign quadrants split by the diagonals — and flag it as inference. And the original's convergence behavior, which the reconstruction's authors themselves identify as problematic, we replace with an explicit stopping rule: iteration ends when the forecast median moves less than half a continuum unit for two consecutive rounds, or at a hard cap, with the rule that fired recorded in every output. Where an ambiguity admitted multiple readings, we adopted the one that reproduces the published replication and documented the choice — replication as specification. One such choice was reached the hard way: an initial reading of the proposal-selection rule produced a systematic downward drift in outcomes, detectable only because the replication target existed to fail against; the corrected ordering (most enforceable proposal first, least movement as tie-break) is itself a small empirical finding about what the original must have done.

The gate for all of this is the model's canonical worked example: the European emission-standards case, whose input table we transcribed twice independently and diffed before use. Our implementation converges in three rounds to a settlement median of 9.530, within half a unit of both the reconstruction's reported steady state (9.9) and the original's stabilized prediction (9.05), with intermediate round means matching the published trajectory to within a quarter unit. Two properties of this result matter beyond the headline number. First, the coalition structure is preserved — the high-position bloc, the low cluster, and the isolated small actor remain distinct — so the median match is not an artifact of positions collapsing to a point, a failure mode that can counterfeit convergence. Second, one residual deviation exists: the reconstruction reports a first-round transient our implementation does not reproduce, which we attribute to the figure-only octant boundaries and quantify rather than conceal; the acceptance tolerance was fixed in advance and never loosened to pass. The replication test remains in the codebase as a permanent regression gate: the model cannot drift from its published behavior without a test failing.

3. The fair fight: evaluation on expert-coded legislative bargaining

A model that claims to forecast collective decisions should be examined on the one benchmark built for exactly that claim. The Decision-Making in the European Union datasets record real legislative controversies as this model family represents them: for each contested issue, a continuum of outcomes, expert-coded actor positions and saliences, and — decisively — the outcome actually adopted. The data were assembled independently of any one model, have anchored two decades of published model comparisons (Thomson et al. 2006; Schneider et al. 2010), and are immune to hindsight contamination because the inputs were coded by domain experts against the controversy itself, not reconstructed after resolution. Our evaluation set comprises 351 scoreable issues; the release we use is the combined third-wave dataset, which already merges all three collection waves — DEU I, II, and III, spanning 1999 to 2019 (Arregui & Perarnaud 2022) — so the benchmark's size is fixed by the field's twenty-year record, not by our sampling, and there is no earlier wave left to add. The error metric is the absolute distance between a model's forecast and the adopted outcome on the issue's own scale; and the decision criterion was pre-registered before any result was computed: the challenge model must beat two naive baselines — the capability-and-salience-weighted mean of positions, and the position of the median actor — on mean absolute error, or the gate fails.

The first evaluation handicapped the model, and we report it because the handicap is instructive. The DEU datasets record no capability as a coded variable: the exchange-model framework's third input, actor power, has always been supplied exogenously — in practice as Council voting weights — rather

than elicited into the dataset, so “capability” is a modelling choice the analyst brings, not a column the experts filled. Absent a sourced table, all actors initially received equal capability — an assumption that amputates the model’s defining input, power asymmetry, while quietly converting the rival baseline into the salience-weighted mean, the specification the literature already identifies as formidable (Achen 2006). Under these conditions the challenge model failed the gate (28.31), edging the median baseline but losing clearly to the weighted mean (23.64). Two objections demanded a rematch. First, capabilities: we constructed a sourced capability table from Council voting weights under the treaty regime governing each issue, with a documented treatment of the Commission and Parliament following the dataset’s own conventions, every number cited. Second, the reference point: the original evaluation’s configuration discarded the status quo entirely, yet DEU records a reference point on 261 issues, and the model family’s own theory assigns it a role. The fair fight restored both, with the gate re-registered in advance: the fully equipped challenge model against the equally equipped weighted mean.

It failed again. Restoring the model’s genuine inputs — the power asymmetries and status-quo anchors its theory demands — did not close the gap to the simple aggregate (26.83 / 38.51 vs 22.99 / 29.77), each figure pair reporting mean absolute error / root-mean-square error, and the paired difference in mean absolute error is bounded away from zero by its bootstrap interval (+3.84, 95% CI [+1.66, +6.01]). We state this plainly and do not soften it: the pre-registered criterion was mean absolute error, and on mean absolute error the challenge model loses. The anatomy of the failure is more informative than its size. The challenge model’s errors are bimodal: its worst cases are stampedes to the extremes of the scale, forecasting one pole while the outcome sat at the other. The mechanism is visible in the model’s own logic — the weighted-median dynamic allows a numerous but individually weak bloc to capture the pivotal position and drag the converged forecast to a pole, precisely the configuration in which a smooth weighted average cannot be dragged; and several of the worst misses are status-quo reversions, outcomes at the reference point that the challenge dynamics, built to simulate movement, structurally under-predict. These are not implementation artifacts: the same error signature motivates the successor search of Section 4 and reappears, as a diagnostic warning, in live use of the released software.

The result is specific to the loss function

The pre-registered gate is settled; a reviewer’s question sharpened the reading, and we report the answer as what it is — post-hoc and responsive to review, not a pre-registered result. Under mean absolute error the compromise mean wins. Under other criteria the picture changes. On *median* absolute error (20.00 vs 17.54) the paired difference is smaller and its bootstrap interval spans zero (+2.46, 95% CI [-2.34, +4.17]) — by that measure the two are indistinguishable. On the hit rate — the fraction of issues forecast within five, ten, or twenty scale units of the outcome — the challenge model is *ahead* at the tight thresholds: it is more often nearly exactly right. On the continuous ranked probability score, computed against the capability-and-salience-weighted distribution of positions, the two are essentially tied (18.38 vs 18.14). And issue by issue the margin is narrow rather than a rout: the compromise mean is nearer on more issues than not, but the split does not reach significance under a sign test (157 vs 194; sign test $p=0.055$). The pre-registered conclusion stands on the metric it was registered on; it does not generalise to every loss a user might care about.

The reconciling picture is a bias–variance trade, and it replaces “the challenge model loses” as this paper’s characterisation of the result. The compromise mean is a low-variance, biased estimator: it never stampedes, so its errors are moderate and clustered — exactly what a squared or absolute loss

rewards. The challenge dynamics are the opposite, a higher-variance, lower-bias estimator: they more often land squarely on the outcome and more often miss by a pole, a bimodal profile whose extra exact hits are bought with extra catastrophes. Under absolute-error loss the trade is bad, because the loss is dominated by the fat tail; under a hit-rate loss, which only asks whether the forecast was close, the trade is favourable. The challenge model does not lose everywhere: it trades bias for variance, and whether that trade is worth making is a property of the user's loss function, not of the model alone. What it does not do, under any loss we tested, is *beat* the mean on the criterion we pre-registered.

Two comparisons discipline the interpretation. Our error magnitudes sit squarely in the regime the published literature reports for this model family on these data — the reconstruction-era tests of the original model, and its inventor's own later comparisons (Bueno de Mesquita 2011), place it in the same range and losing to the same baseline (Old Model 21.5/28.2 vs weighted mean 11.8/19.4, weighted mean does as well or better than complex models), though scoring conventions differ enough that we flag these as context rather than direct comparison. Landing where the genuine article lands is, simultaneously, the strongest external evidence for our replication's fidelity and the least flattering possible frame for the model itself: the failure we report is not a defect of our copy but a property of the original. What the failure does *not* yet establish is why the simple aggregate is so hard to beat. Section 4 reports our attempt to beat it; Section 5 reports the answer.

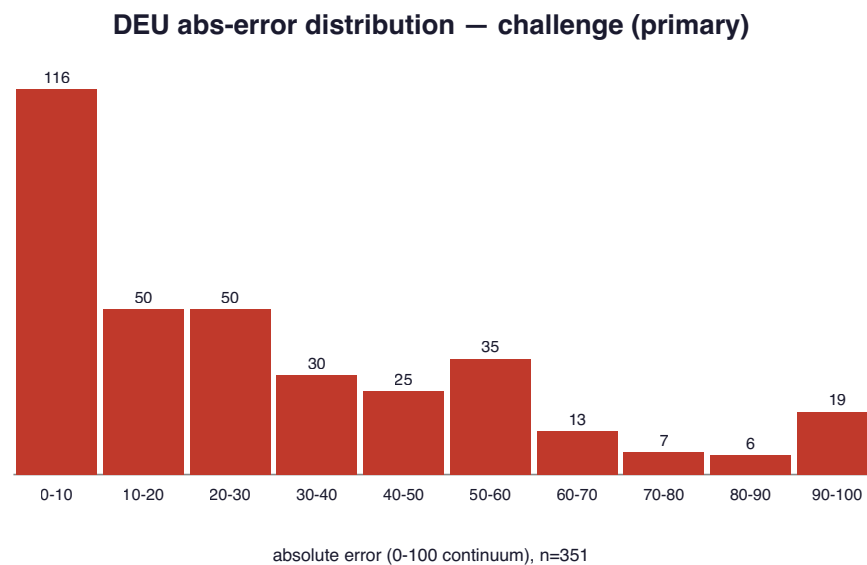


Figure 1. Absolute-error distribution of the challenge model on the 351 DEU issues — bimodal, with pole-to-pole misses.

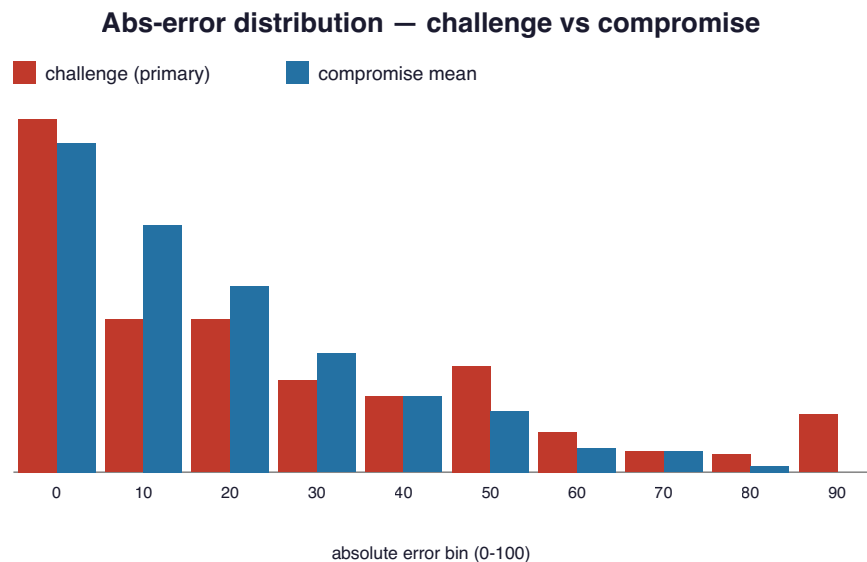


Figure 2. Absolute-error distributions compared: challenge (primary) vs the compromise weighted mean.

4. A pre-registered successor search

The fair fight established that the challenge model does not beat the weighted mean; it did not establish that nothing does. The mean’s dominance could reflect a genuine ceiling, or merely the absence of a serious challenger. We therefore mounted the search — under rules designed so that failure would be informative rather than deniable.

The rules are the section’s first contribution. Before any candidate model was specified, the 351 issues were partitioned by seeded draw into training, development, and test splits of 140, 105, and 106 issues, and the partition was committed to version control. Only afterwards was any model code written. Because the repository’s history is append-only and its commits are content-addressed, the ordering itself certifies the protocol: the commit containing the frozen partition precedes the commits containing the candidates, and no retrospective edit can reverse that order without detection. Model selection used the development split only; feature standardization used training statistics only; the test split was scored exactly once, at the end, with paired bootstrap confidence intervals. This is pre-registration implemented in the researcher’s own working medium — cheaper than a registry, self-enforcing, and independently auditable by anyone with the repository. We offer the pattern as a general method for computational studies.

Two candidates were fielded first, each aimed at a failure mode documented in Section 3. The *status-quo gravity* model addresses reference-point blindness: it forecasts a convex combination of the weighted mean and the reference point, with the mixing weight a learned function of structural features — the decision rule in force, the concentration of capability, the distance between the reference point and the crowd. The *regime-aware settlement* model addresses the bimodal stampedes: it posits that controversies come in kinds — consensual bargains, coercive contests, deadlocks — estimates regime probabilities from observable structure, and blends the compromise, challenge, and status-quo predictions accordingly. Both are structurally motivated, modestly parameterized, and fitted strictly under the protocol.

Both failed. On the untouched test split, the gravity model scored worse than the compromise baseline by 0.83 scale units, with a confidence interval spanning zero (22.09 vs 21.26); the regime model scored worse by 0.48, likewise spanning zero (21.57 vs 21.09). Neither candidate is

distinguishable from the mean; neither improves on it. The two candidates are held to different compromise baselines — 21.26 for gravity, 21.09 for regime — because the gravity model is defined only on the issues that carry a reference point and is scored on that subset, while the regime model is scored on the full test split; the gap is a subset difference, not a difference in data. The more informative result is *how* the regime model failed: granted complete freedom to weight its three components, the fitted model collapsed — assigning roughly five-sixths of its weight to the compromise component and treating the challenge and status-quo terms as noise to be minimized. The search, in other words, converged on the baseline it was built to beat. We read this as evidence that the structural features available in these data carry no exploitable regime signal *within* the domain: if decision regimes differ, they differ between domains — cooperative lawmaking versus coercive crisis — not measurably within this one. The regime hypothesis thus survives only in the cross-domain form that Section 7’s apparatus exists to test, and in no weaker form.

Two fitted blends test only one corner of the design space, so we widened the search to the discipline’s other traditions, each fielded under the identical gate — parameter-free, or with its single constant fixed in advance, and scored on the same held-out test split. *Quantal-response equilibrium*, which replaces the challenge model’s hard best-response with a logit choice at a rationality parameter fixed before any run, moved the error the wrong way (24.63 vs 21.09, $\Delta +3.54$, CI [-0.29, +7.40]). The two *axiomatic bargaining* solutions — Nash’s product-maximizing point and the Kalai-Smorodinsky proportional one, each taking the reference point as the disagreement outcome — did worse still (46.59 vs 21.26, $\Delta +25.33$, CI [+17.64, +32.90]), (32.41 vs 21.26, $\Delta +11.16$, CI [+4.36, +17.79]); we conjecture that the cooperative idealization sits poorly against outcomes coded at the poles, though we do not test the conjecture. *Probabilistic Condorcet* — the method the open-source successor KTAB actually ships, turning the pairwise contest into a stationary distribution over winners — came the closest of anything we tried (20.43 vs 21.09, $\Delta -0.66$, CI [-2.26, +1.00]): a nominally lower error than the mean’s, and the only method to post one, but with a confidence interval that straddles zero and so cannot be told apart from it. Counting the incumbent, the tournament now spans 8 distinct solution concepts — expected-utility bargaining, quantal response, axiomatic bargaining in two forms, probabilistic voting, two fitted structural mixtures, and the influence-weighted mean itself — all measured on the same held-out issues, and the mean has outlasted every one. The one method that drew level did so within the noise.

That comparison carries a limitation we state rather than bury. The test split was committed to version control before the two R1 candidates and scored once for them, as pre-registered; the four Phase C methods were then scored on the same split afterwards. None of the four was tuned on it — each is parameter-free or fixed a priori — but the split was no longer virgin. Counting the two R1 candidates and the four Phase C methods, six models have now been measured against the mean on this one held-out split, and reusing a test set that way inflates the chance of a lone nominal winner: under the null that no method beats the mean, seeing one of six point estimates fall below it is about what chance alone produces, not a discovery. Probabilistic Condorcet is that lone nominal winner, and it does not survive being counted honestly. Its margin already lies inside its own 95% interval, and a multiplicity correction only sharpens the verdict — a Bonferroni threshold of 0.05/6 demands far more than an interval grazing zero. We read PCE as the coin that came up heads once in six tosses; the finding stands, and the split reuse is the caveat, not a caveat-shaped escape from it.

4.1 The operator, not the dynamics

If seven distinct challengers cannot separate from a simple average, the natural question is what the average is doing that the mechanisms are not. We think the answer is *smoothing*, and two claims are worth keeping apart. The first is mathematical and holds by construction: the challenge model reports, as its forecast, the capability-weighted *median* of the settled positions, and a weighted median is a discontinuous function of its inputs — it snaps from one actor’s position to another’s as the pivotal vote changes and ignores the rest — whereas a weighted mean is continuous and shifts smoothly with every input. That much is not an empirical finding; it is what the two operators are. The second claim is empirical: that discontinuity is not merely formal but produces measurable insensitivity — a *median lock*, in which the forecast stops responding to its own inputs. Here the evidence is thin and we say so. The benchmark’s 351 issues are point estimates with no ranges to perturb, so the lock can be probed only on the two live geopolitical games the engine now carries under pre-registration — two cases, not a benchmark. On both, a one-at-a-time sensitivity sweep finds that most parameters do not move the forecast at all.

Quantal response is a pointed test of whether that lock lives in the bargaining dynamics or in the operator beneath them, because it softens precisely the discrete step — which offer a mover accepts — that the hard model resolves by fiat. It does not help. Re-running the sweep under the softened dynamics leaves the count of dead parameters essentially unchanged — (18/27 → 17/27) on the first game and (13/20 → 13/20) on the second — and, the diagnostic’s real tell, it *tightens* the forecast interval rather than widening it: the 80% band contracts on both games (44.65 → 25.02), (32.71 → 22.28). That combination is itself a warning, and one the paper’s own metrics act on: on the benchmark the softened model was *less* accurate, not more, so a narrower interval around a no-better forecast is overconfidence — a forecast grown more precise without growing more correct — which is exactly the failure a proper scoring rule penalises. Softening the choices damps the excursions that did occur without ever freeing the median that was stuck. On these two games, then, the lock tracks the operator rather than the dynamics above it: a weighted median is pinned by whichever coalition owns the pivot, and smoothing the negotiation that feeds it does not, in these cases, unpin it.

That points to the fix in kind, without claiming it is the only one. The weighted *mean* is the simplest operator that is continuous in every input, moves whenever any actor moves, and cannot lock — and it is the one that wins here. It is not the only such aggregator: other smooth functions of the same inputs exist, and we have not tested them. But the mean is not a rival mechanism that merely happens to score well; it is the kind of smoothing the median lacks and, being discontinuous, structurally cannot supply. The tournament then carries a single moral: the predictive content of these games lives in an average, and a solution concept that refines the *dynamics* while keeping the median *operator* inherits the operator’s insensitivity and forfeits the gain.

A note on provenance, reported because methods sections should describe what actually occurred. The two fitted candidates were proposed by a large language model working from Section 3’s failure diagnostics; the other concepts in the tournament — the challenge model and the four further methods — are standard constructions drawn from the game-theory literature, not machine-generated. The implementation itself — the solver, the search harness, and the scoring code — was likewise developed with AI coding assistance, under the same pre-registered discipline: the train/development/test split and every acceptance threshold were committed to version control before any model or its scoring code was written, so the repository’s history certifies that no code, whatever authored it, could consult held-out data before its verdict was fixed. In every case the protocol, execution, and verdicts belong entirely to the deterministic apparatus. The division — hypotheses

proposed, from whatever source, and a pre-registered harness disposing of them — functioned exactly as intended, here by killing every proposal. We record the pattern without advocacy, as one datum on how candidate models can be subjected to conventional falsification standards.

Seven attacks on the mean — a celebrated mechanism restored to full strength, its quantal-response softening, two axiomatic bargaining solutions, the probabilistic-Condorcet method of the field’s own successor library, and two fitted blends on held-out data — have now failed. The remaining question is no longer *which* model beats the weighted mean on these data, but whether *any* model can. Section 5 gives the answer.

Successor leaderboard — TEST scored once (R1)				
Candidate	TEST MAE	comp MAE	delta [95% CI]	beats?
Candidate A — status-quo gravity	22.09	21.26	+0.83 [-0.15, +1.91]	no
Candidate B — regime-aware settlement	21.57	21.09	+0.48 [-0.69, +1.76]	no

pre-registered split train 140 / dev 105 / TEST 106 (seed 20260721); no candidate beats the compromise mean.

Figure 3. Successor-search leaderboard — neither candidate beats the compromise mean on the untouched TEST split.

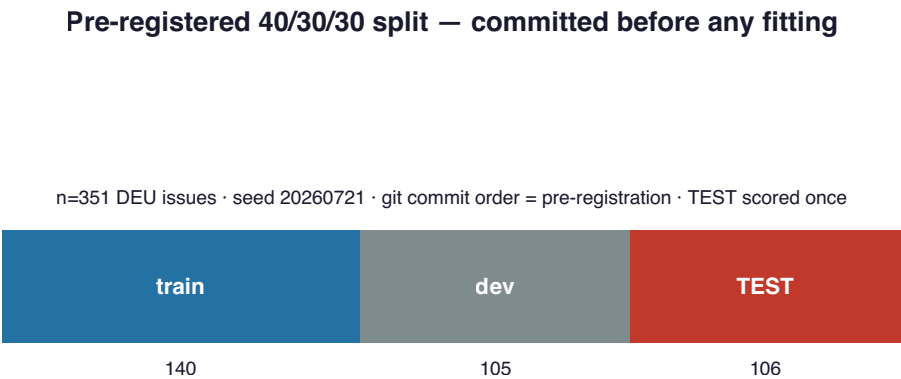


Figure 4. The pre-registered 40/30/30 train/dev/TEST split, committed before fitting.

5. The noise floor: locating the predictability ceiling

Seven failures admit two readings. Either the challengers were the wrong ones — and a better model waits undiscovered — or there is no further signal to find, and every elaborate mechanism was mining an exhausted vein. These readings are distinguishable, because the second makes a testable claim: if the classic input set contains predictive signal beyond the weighted mean, a sufficiently flexible learner, honestly cross-validated, should recover at least some of it. We built that learner as a deliberate ceiling probe — a diagnostic, not a product — applying to political forecasting the predictability-ceiling method a mass scientific collaboration recently used to bound the forecastability of the life course (Salganik et al. 2020). Eighteen features were derived from the raw inputs, including position summaries, dispersion measures, and salience-weighted quantiles of the

actor distribution; a family of models spanning linear to kernel ridge regression was fitted under seeded five-fold cross-validation with within-family selection. The clean form of the test fits the learner not to the outcome but to the *residual* the weighted mean leaves behind — the part of each outcome the mean does not already explain — and asks how much of that residual any flexible model can recover.

It recovers nothing. The cross-validated coefficient of determination on the residual is indistinguishable from zero, its bootstrap interval spanning both signs (-0.029, 95% CI [-0.093, 0.019]): a model granted every freedom — nonlinearity, interactions, eighteen engineered views of the data — extracts no signal beyond the influence-weighted average. The design earns its conclusion in the conservative direction. Because the weighted mean is itself one of the features, fitting the *outcome* would let ridge shrinkage bias the mean's own coefficient away from one and flatter the flexible model; fitting the *residual* shrinks toward zero — which is exactly the baseline — so any recovered signal would be real, and the offsetting optimism of in-sample hyperparameter selection cuts the other way and still finds nothing. The benchmark's resolution is finite in a way worth stating precisely: at 351 issues the smallest paired difference in mean absolute error detectable at conventional power is roughly three scale units (3.04) — a detection floor that is a property of the benchmark rather than of our analysis, and because the 351 issues already combine all three DEU waves, the benchmark cannot be enlarged, so effects below that floor are undetectable here and would remain so with more of the same data, of which there is none. The claim now rests on two independent lines of evidence that meet at the same floor. From above, the residual probe recovers nothing the mean has left. From the side, a tournament of 8 distinct solution concepts spanning the field's major traditions — expected-utility bargaining and its quantal-response softening, the Nash and Kalai-Smorodinsky bargaining solutions, probabilistic Condorcet, two fitted structural mixtures, and the weighted mean itself — ranks them so that the mean finishes first and no rival separates from it. Sections 3 and 4 are thereby reframed: the challenge model, the bargaining solutions, the Condorcet method, and the fitted blends were not defeated by a superior competitor. They were competing for signal that does not exist.

Read positively, this is a substantive claim about politics, not merely about method: once an analyst knows who is at the table, what each actor wants, how much power each holds, and how much each cares, the influence-weighted average of their positions extracts everything those facts contain about the outcome of consensual bargaining. The residual error — some twenty-three scale units — divides between measurement granularity in the outcome coding and genuine historical contingency, the portion of politics that is simply not written in positions, salience, and capability; we take up the division in Section 9. One scope condition belongs here rather than with the general limitations, because it qualifies the ceiling directly: this is a ceiling for **mean-error losses** — the mean absolute and squared errors on which the weighted mean is the optimal point estimator — not a ceiling for every loss. Section 3 showed the challenge dynamics ahead on tight hit rates; a user who scores by near-exact hits rather than by average distance is not at the mean's ceiling but on the other side of a bias-variance trade. Within mean-error losses, and within the domain and input set Section 9 delimits, the search for a cleverer aggregation of the classic inputs is, on this evidence, complete — a domain-specific instance of the broader forecasting finding that added complexity rarely earns its keep against a well-specified simple method (Green & Armstrong 2015).

Which leaves the lineage's history in need of a different explanation. If any sane aggregation of well-elicited inputs reaches the ceiling, then the celebrated performances of Policon and its descendants cannot have come from their solution dynamics. Section 6 says where they came from.

6. Structure, not magic: reinterpreting the lineage's record

The strangest sentence in the declassified record has waited forty years for an explanation. Feder's evaluation reports not merely that Policon's forecasts were frequently accurate, but that the model outperformed the very analysts who supplied its inputs — the same experts whose judgments of position, salience, and capability were the model's only contact with reality somehow produced better forecasts through the model than they produced themselves. The natural reading, and the one the lineage's mystique rested on, is that the solution dynamics added something: that the rounds of simulated challenge and coalition were extracting insight the analysts could not. Our results make that reading unnecessary. If the weighted mean of well-elicited inputs sits at the extractable-signal ceiling — and if the elaborate dynamics merely match it, in our tests and in the published record alike — then the equations contributed approximately nothing, and the performance must have come from somewhere else in the pipeline.

It came, we argue, from the step everyone looked past: elicitation. Requiring an expert to commit to numbers — *this* actor at *this* position, caring *this* much, wielding *this* much power — is a technology in its own right. It strips narrative hedging; it forces every actor into a common metric; it prevents the vivid case from being counted twice and the dull one from being forgotten; and it hands the final synthesis to an aggregator that weighs many small factors with a consistency human judgment cannot sustain. Framed this way, the finding that startled the CIA is a familiar one wearing a uniform. That mechanical combination of structured inputs beats the unaided judgment of the same experts is among the most replicated results in decision science — from the clinical-versus-statistical prediction literature (Meehl 1954) through the demonstration that even *improperly* weighted linear models outperform the experts who feed them (Dawes 1979), to the structured-judgment findings of the forecasting-tournament era (Tetlock 2005). Policon, on this reading, was a classified rediscovery of the improper-linear-models result: the analysts' numbers plus almost any consistent aggregation would have beaten the analysts' narratives. The convergent evidence runs the other direction too — game-theoretic expertise *without* structured protocol has been found to forecast conflict no better than chance (Green 2002; 2005) — so the operative ingredient across both literatures is neither the theory nor the theorist. It is the structure. Not magic.

This reinterpretation also explains a commercial puzzle the mystique never could: why four decades of unreplicability never damaged the lineage's business. If the value had lived in secret equations, clients were buying a black box that independent tests now show adds nothing. But if the value lived in the elicitation practice — the disciplined interviews, the bounded questions, the conversion of expertise into structured inputs — then the consultants were delivering it in person on every engagement, equations or no equations. The secret stayed safe because the secret was not where anyone was looking.

A second mechanism, cognitive rather than commercial, is one the loss-function result of Section 3 now lets us name. The challenge dynamics do not produce uniformly mediocre forecasts; they produce a *bimodal* profile — an unusual share of near-exact hits bought with an unusual share of pole-to-pole misses, which is why they lead on tight hit rates even as they lose on average error. That profile is the raw material of a legend. Human judgment weights evidence by availability, not by average (Tversky & Kahneman 1973): the vivid case — the forecast that landed on the number while every analyst hedged — is recalled, retold, and folded into the track record, while the catastrophic miss is forgotten, rationalised as a genuine surprise, or quietly dropped. A model whose errors were moderate and clustered, like the weighted mean's, generates no such stories; a model that is

sometimes uncannily exact generates them constantly. The lineage’s reputation for accuracy, on this reading, was neither manufactured nor quite earned — it was the predictable product of a high-variance forecaster meeting a memory that samples on the hits, while the structured elicitation underneath did the quiet, unmemorable work of being reliably close.

We state the claim’s epistemic status precisely, because this section carries the paper’s title. The reinterpretation is an inference to the best explanation, not a direct test: we have not run elicitation experiments, and we do not claim to have proven that structured inputs alone generated the historical record — only that, given the ceiling result, the solution dynamics are unnecessary to explain it, and the structured-judgment literature supplies a mechanism that is sufficient, parsimonious, and independently replicated. The direct test is specifiable — compare aggregators on the lineage’s own published, ex-ante-coded applications — and Section 7’s case library is its beginning. One modern implication we note and defer: the elicitation step, historically the expensive and human-limited part of the pipeline, is now partially automatable with language models producing evidence-cited input tables for expert review; our released software implements this, and it relocates where scrutiny in this model family belongs — from solution concepts, where forty years of attention produced nothing measurable, to input quality and provenance, where everything measurable lives.

7. A verifiable case library for the open question

The question this paper declines to answer — whether the challenge dynamics earn their keep in the coercive settings they were designed for — cannot be answered with existing benchmarks: crisis datasets record events and outcomes but not the stakeholder tables this model family consumes. The inputs exist elsewhere. Across four decades, the lineage’s own published applications printed their expert-coded tables — positions, salience, capability, and the analyzed continuum — for cases whose outcomes are now historical fact, many coded genuinely ex ante. We contribute the protocol for turning that scattered record into a benchmark, and its first verified entries.

The protocol has four commitments. A canonical case schema preserves what evaluation requires: the actor table on the source’s own scales; the continuum with its printed qualitative markers, so that outcome coding maps historical events to the *source’s* definitions rather than ours; the publishing study’s own forecast, retained as a scoreable rival; outcome readings at explicitly labeled horizons, with the source’s stated horizon designated primary; and a domain tag, because a case’s evidential weight depends on which question it can address. Transcription is *blind dual-entry*: a first transcription from text extraction, a second performed blind against rendered page images, and a machine diff of the two — with arithmetic checksums exploited wherever the source prints derived columns, so that a mistyped digit fails arithmetic rather than surviving into analysis. Verification is separated from judgment: a clean sixty-row diff can certify digits (60/60 (26 + 34)), but the verified flag flips only on a human’s recorded ratification of the judgment items — outcome codings, horizon mappings — with the ratification quoted in the case file. And verdicts are suppressed by construction: the evaluation harness refuses to report a domain comparison while the library is small, entries are unverified, or cases fall outside the claimed domain, and the coercive verdict is pre-registered to be read once, at eight verified cases, and not before. At writing, the library holds two verified cases — domestic-elite bargaining, valuable as out-of-domain validation and explicitly *not* coercive — and the coercive column of every results table in this paper reads PENDING. We regard publishing the apparatus with its verdict withheld as the point, not a limitation: the alternative literature is one of verdicts published without apparatus.

8. A commit-reveal ledger for live forecasts

The record that opened this paper — ninety percent accuracy, asserted from inside a classified program — exemplifies the general condition of forecasting track records: they are curated memories, unverifiable at the level of the individual claim, revisable in the telling. The determinism built into our implementation permits a stronger standard. Because a forecast record is a pure function of its inputs, configuration, and seed — byte-identical on every regeneration — its SHA-256 digest is a complete commitment to its contents. We therefore publish, in the repository itself, the digest of each forecast *before* its target event resolves; the full record is revealed at grading, when anyone may recompute the digest and confirm that the sealed prediction is the graded one. The repository's append-only history timestamps every commitment; backdating is detectable; and corrections are possible only in public, on top — a property we can attest from experience, having corrected our own ledger's initial hash basis in situ, with the correction documented in the ledger it corrects. The instance is live and has grown: the sealed ledger now spans three pre-registered questions and fourteen records, of which one question and its six records have been graded — the first graded cycle, reported below. The earliest sealed instance is four forecasts on one ongoing geopolitical question — the two solver philosophies of Sections 3–5, each at two input vintages — sealed on their stated dates (34.576, 41.636, 29.407, 39.443) and committed to grading on schedule regardless of what reality returns; the live runs have already exercised the diagnostics of Section 3, the released software flagging its own challenge-model instability in production. The methodological claim is modest and, we think, portable: any deterministic forecasting system can adopt this standard at the cost of one hash per forecast, after which its accuracy claims become falsifiable at the level of the individual prediction — which is precisely the level at which the literature's founding legend was never checkable.

The ledger has since recorded its first *graded* cycle, and we report it here with its result and, more to the point, its limits. The pre-registered question Q-2026-OPEC-SEP asked what collective production adjustment the OPEC+ group would announce for September 2026; its rubric, fixed in writing before the event, maps the announced figure onto the settlement continuum by a published arithmetic rule — an increase of 188 thousand barrels a day settling at 66 on the 0–100 scale under the rubric's committed linear map. On 2 August 2026 the participating countries announced exactly that increase (OPEC Secretariat statement, 2 August 2026). At grading, all 6 sealed records — both input vintages across all three model families — re-verified against their published hashes and were scored under the rubric's own arithmetic, with no discretion applied.

We state the caveats plainly rather than bury them, because a single graded point invites exactly the over-reading this project exists to resist. First, one graded question is no evidence of forecasting accuracy, and we claim none: it lies far below the pre-registered threshold at which our ranking guard permits any model family to be compared, let alone ranked. Second, the announced adjustment was the consensus expectation of oil-market analysts in the weeks before the meeting, so the question had little power to discriminate between forecasters — an easy call graded correctly establishes almost nothing. Third, our primary and secondary metrics disagree about which solver came closer: on absolute error the sourced challenge record edged the compromise mean (2.320 vs 2.566), while on the continuous ranked probability score the ordering reverses and the compromise mean is nearer (1.535 vs 1.611); we report the disagreement rather than resolve it. What the cycle demonstrates is the machinery end to end — sealed commitment, public rubric, cryptographic re-verification, and an

arithmetic grade computed without discretion — while establishing nothing whatever about accuracy, which is precisely the posture we intend at a sample size of one.

9. Limitations

The measurement floor comes first, because it bounds everything quantitative. Three-quarters of DEU outcomes are coded at round multiples of ten, so the residual error against which all models were judged contains an irreducible granularity component, and the ceiling estimate of Section 5 inherits it: our claim locates the ceiling of *these codings*, and some fraction of the twenty-three-unit residual is the coding, not the politics. The worst-case failures (10 issues at abs error up to 100.00) carry the same caveat in miniature — a hundred-point miss presupposes that the original coders' mapping of a messy legislative outcome onto a continuum endpoint is itself beyond dispute, which it is not always. Second, construction choices: DEU records no capability, and our sourced table — Council voting weights by treaty regime, with a documented treatment of the Commission and Parliament — is one defensible reconstruction among several; we tested the model with the most faithful inputs we could assemble, not with every capability theory an advocate might prefer, and every choice is cited precisely so that an advocate can rerun the pipeline under a different one. Third, scope, restated as a boundary rather than rediscovered as a concession: all quantitative results concern cooperative legislative bargaining under the classic input set; other cooperative arenas are plausible but untested extrapolations, richer input universes were not probed by our oracle, and the coercive domain is open by design. Fourth, the apparatus of Sections 7 and 8 is young: a two-case library supports no comparison and is prevented by its own guards from offering one, the eight-case reading threshold is a pre-registered judgment rather than a derivation, and the ledger, at writing, has sealed forecasts and a single graded cycle (Section 8) — enough to exercise the standard end to end, but far too few to be a track record or to support any claim of accuracy. Fifth, we replicated the reconstruction, because the reconstruction is all that publicly exists; if the original model diverged from it in ways its authors never disclosed, our results characterize the only version of the model that can be tested — and we note, deliberately, that any defense of the original resting on undisclosed differences concedes the unfalsifiability this paper identifies as the deeper problem. Finally, the reinterpretation of Section 6 rests on inference, not experiment: the elicitation step itself — the mechanism we credit — was taken as given throughout, and its direct evaluation is the clearest item of future work this paper leaves open.

10. Conclusion

We set out to close an anomaly: a celebrated forecasting model, credited with remarkable accuracy inside a classified program, that no one could independently run. Closing it took four steps, each smaller than the legend. An open implementation reproduced the published behavior. A fair evaluation on the canonical benchmark — the model's real inputs restored — found it beaten by the weighted mean of its own inputs. A pre-registered search for a better successor produced two candidates that failed on held-out data, one by collapsing onto the baseline it was built to beat. And a ceiling probe found no signal left to extract: on these data, under these inputs, the simple aggregate is not merely hard to beat; there is nothing beyond it to find. What remains of the legend, we argued, is real but relocated — the discipline of structured elicitation, a technology decision science has validated for seventy years, performing exactly as that literature predicts while the equations above it added nothing measurable.

Three recommendations follow for the field. Retire the solution-concept question for cooperative bargaining; the evidence, ours and the literature's, says it is settled. Redirect scrutiny to where the signal enters — elicitation quality and input provenance — including the new question of language-model-assisted elicitation, which our released software implements and which will need exactly the audit standards this domain has lacked. And adopt verifiability as infrastructure, not aspiration: pre-registration in version control and commit-reveal sealing cost almost nothing, and they would have made the last forty years of this literature checkable. One question stays deliberately open, with its apparatus published, its threshold pre-registered, and its verdict withheld until the evidence exists: whether the challenge dynamics earn their keep where coercion, not consensus, sets the rules. The founding claim of this lineage could never be checked. Every claim in this paper can be. That difference, more than any model or any verdict, is what we hope survives.

Declarations

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Competing interests. The author declares no competing interests.

Data availability. The code and all derived artifacts — the backtest tables, the sealed forecast ledger, the evidence file, and the figures — are in the public repository at <https://github.com/hamzagulo7/schelling>. The DEU III dataset is not redistributed here and must be obtained from its open deposit (doi:10.34810/data53, CC BY 4.0) and placed at `data/deu/`. With that dataset in place, the fair-fight, successor-search, and predictability-ceiling results of Sections 3–5 regenerate by a single command; without it, the replication reference case (Section 2), the case library (Section 7), and the sealed forecast ledger (Section 8) still regenerate from committed artifacts, while the DEU-dependent tables are reported as open questions.

AI-assistance disclosure. Both hypothesis generation and implementation used AI assistance under pre-registered constraints. The two fitted successor candidates were proposed by a large language model (Section 4), and the implementation — the solver, the search harness, and the scoring code — was developed with AI coding assistance. In every case the train/development/test split and all acceptance thresholds were committed to version control before any model or scoring code was written, so the repository's history certifies that no code, whatever its authorship, could consult held-out data before its verdict was fixed; the protocols, execution, and verdicts belong entirely to the deterministic apparatus.

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Appendix: provenance of cited figures

Every figure the paper cites is regenerated by `schelling paper-evidence` from the listed artifact; the E-tag keys each value to its source and provenance stamp.

E-tag	Value	Source	Provenance
E-REPL-MEDIAN	9.530	tests/fixtures/ emission_standards.json	b5ad732
E-DEU-N	351	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-DEU-GATE	FAILED	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-R1-gravity	22.09 vs 21.26	src/schelling/backtest/ deu3_split.json	sha256:0d75f0d2f3a9
E-R1-regime	21.57 vs 21.09	src/schelling/backtest/ deu3_split.json	sha256:0d75f0d2f3a9
E-ORACLE-GAP	-0.84	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-CHINA-VERIFIED	True	data/coercive-cases/ ktab-china-2014.json	ca65c05
E-LEDGER	34.576, 41.636, 29.407, 39.443	FORECASTS.md	family: E-LEDGER- challenge-v1, E- LEDGER-compromise- v1, E-LEDGER- challenge-v2, E- LEDGER-compromise-v2
E-DEU-MAE-r1	28.31	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-BASE-WMEAN-r1	23.64	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-METHOD-capabilities	treaty-regime Council power (pre-Nice / Nice / Lisbon), rescaled strongest=100	src/schelling/backtest/ capability.py	c3de672
E-METHOD- challenge_rp	26.83 / 38.51	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-METHOD- baseline_wmean	22.99 / 29.77	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-REVIEW-MAE-CI	+3.84, 95% CI [+1.66, +6.01]	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-REVIEW-MEDIAN- AE	20.00 vs 17.54	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-REVIEW-MEDIAN-CI	+2.46, 95% CI [-2.34, +4.17]	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-REVIEW-HITS	err	<=5/10/20, challenge vs compromise	<=5: 0.29 vs 0.17; <=10: 0.41 vs 0.31; <=20: 0.55 vs 0.54
E-REVIEW-CRPS	18.38 vs 18.14	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-REVIEW-WINS	157 vs 194; sign test p=0.055	data/deu/ Dataset_DEU_III.csv	sha256:0d75f0d2f3a9

E-tag	Value	Source	Provenance
E-CTX-bdm2011	Old Model 21.5/28.2 vs weighted mean 11.8/19.4	BACKTEST.md	c71f138
E-CTX-achen2006	weighted mean does as well or better than complex models	BACKTEST.md	c71f138
E-R1-SPLIT	140 / 105 / 106	src/schelling/backtest/deu3_split.json	2b1e855
E-PHASEC-challenge-qre	24.63 vs 21.09, Δ +3.54, CI [-0.29, +7.40]	src/schelling/backtest/deu3_split.json	sha256:0d75f0d2f3a9
E-PHASEC-nash	46.59 vs 21.26, Δ +25.33, CI [+17.64, +32.90]	src/schelling/backtest/deu3_split.json	sha256:0d75f0d2f3a9
E-PHASEC-nash-ks	32.41 vs 21.26, Δ +11.16, CI [+4.36, +17.79]	src/schelling/backtest/deu3_split.json	sha256:0d75f0d2f3a9
E-PHASEC-pce	20.43 vs 21.09, Δ -0.66, CI [-2.26, +1.00]	src/schelling/backtest/deu3_split.json	sha256:0d75f0d2f3a9
E-PHASEC-COUNT	8	src/schelling/backtest/deu3_split.json	sha256:0d75f0d2f3a9
E-QRE-ZEROSWING-usiran	18/27 $\rightarrow$ 17/27	analyses/usiran/usiran-v2.json	sha256:8868b82c14e5
E-QRE-ZEROSWING-iaea	13/20 $\rightarrow$ 13/20	analyses/iaea/iaea.json	sha256:afa36056cc44
E-QRE-CI8o-usiran	44.65 $\rightarrow$ 25.02	analyses/usiran/usiran-v2.json	sha256:8868b82c14e5
E-QRE-CI8o-iaea	32.71 $\rightarrow$ 22.28	analyses/iaea/iaea.json	sha256:afa36056cc44
E-REVIEW-RESIDUAL-R2	-0.029, 95% CI [-0.093, 0.019]	data/deu/Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-REVIEW-MDE	3.04	data/deu/Dataset_DEU_III.csv	sha256:0d75f0d2f3a9
E-CHINA-ROWS	60/60 (26 + 34)	data/coercive-cases/ktab-china-2014.json	ca65c05
E-DOMAIN-VERDICT	cooperative: compromise mean wins; coercive: PENDING	BACKTEST.md	c71f138
E-GRADE-RAW	188	FORECASTS.md	cc6124d
E-GRADE-ACTUAL	66	FORECASTS.md	cc6124d
E-GRADE-NSCORED	6	FORECASTS.md	cc6124d
E-GRADE-SOLVER-AE	2.320 vs 2.566	FORECASTS.md	cc6124d
E-GRADE-SOLVER-CRPS	1.535 vs 1.611	FORECASTS.md	cc6124d
E-WORST	10 issues at abs error up to 100.00	data/deu/Dataset_DEU_III.csv	sha256:0d75f0d2f3a9